LESSON 12.1
FINDING SURFACE AREA OF RIGHT CIRCULAR CYLINDERS USING NETS

LESSON INTRODUCTION

MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

LEARNING TARGETS

- I can draw the net of a right circular cylinder.
- I can find the surface area of a right circular cylinder using the figure's net.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- What does surface area represent in a cylinder?
- What is the relationship between a figure's net and the three-dimensional figure?
- How is a figure's net used to find the surface area of a right circular cylinder?

LESSON NARRATIVE

The focus of this lesson is to discover how to find the surface area of a right circular cylinder using the figure's net. Students will make connections between the circumference of the base and the width of the lateral area of a cylinder. Students will derive the formula of surface area by finding the areas of the circular bases and the area of the lateral area. Students will draw nets and find both lateral and surface area for real-world objects.

COMPONENT SUMMARY

12.1.1 WARM-UP *(~5 MIN.)*

Justify which figure doesn't belong

12.1.2 GUIDED INSTRUCTION *(10 MIN.)*

Label the net of a right circular cylinder

12.1.3 EXPLORATION *(15 MIN.)*

Relate the circumference of the base of a cylinder to its lateral area

12.1.4 GUIDED INSTRUCTION *(10 MIN.)*

Find surface area using the net image

12.1.5 WRAP-UP *(~5 MIN.)*

Ticket out the door

RESOURCES

GLOSSARY

Key Terms:

- Cylinder
- Lateral area
- Surface area of a right cylinder

Additional Terms: N/A

MATERIALS

- Toilet paper rolls or index cards
- GeoGebra Applet
- Calculators
- (Optional) String

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may substitute diameter for radius when given the diameter in a model or word problem.
- Students may confuse surface area and volume, or students may interpret surface area as the area of just one surface (i.e., only the base of the cylinder, or only the 'rectangle' formed by the net).
- Students may struggle connecting the circumference of the base to the width of the rectangle that is the lateral area of the cylinder.

THINGS TO CONSIDER

- It may be helpful to provide students with a cutout of a net of a cylinder so they can make the connections between the net image and the cylinder.
- It may be helpful to have string to show students the connection between the circumference of the base to the width of the lateral area. Students often consider anything that uses pi must remain in circular form.
- If time is an issue, consider having students complete question 2 from 12.1.4 Guided Instruction for homework.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

12.1.2 Guided Instruction provides students with the opportunity to discuss the net of a right circular cylinder. Consider applying this routine for questions 1, 2, and 3. Students should answer each problem individually, then share their answers in a low-stakes environment prior to sharing with the entire classroom.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR1.1 Participate in Effortful Learnin

Students will be working on identifying the net of right circular cylinders. As students are working, they will need to ask clarifying questions to complete 12.1.3 Exploration. Support students with scaffolding questions as they discover this new concept.

DIFFERENTIATE INSTRUCTION

12.1.3 Exploration provides students with visual and kinesthetic entry-points as they use a toilet paper roll to model the lateral area of a right circular cylinder. Alternatively, students can use index cards or sheets of paper (construction paper works best).

12.1.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

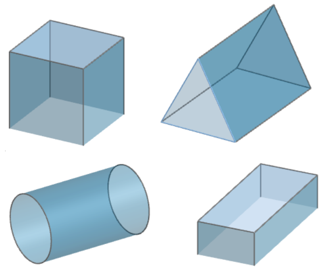
Consider positioning students to discuss question prompts rather than writing their answers. This entry point might engage auditory learners. Use sentence stems to encourage conversation, such as "One detail that I noticed is…" or "This figure stands out to me because…".

QUESTIONING

Challenge students to explain how each figure does not belong to the group.

12.1.1 Warm-Up: Which One Doesn't Belong?

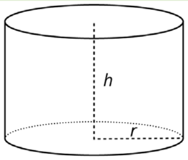
1. Circle the figure that does not belong.



2. Explain why that figure does not belong.
   Answers will vary.

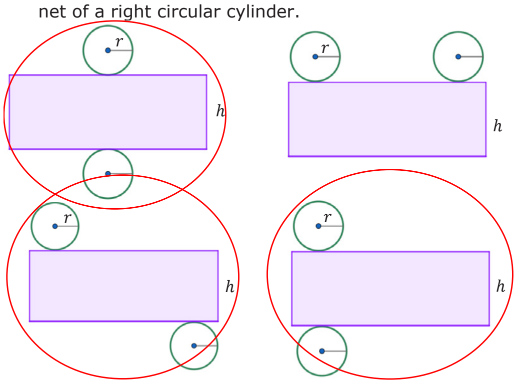
12.1.2 Guided Instruction: Drawing the Net of a Right Circular Cylinder

A ***cylinder*** is a figure containing two congruent, parallel, circular bases whose edges are connected by a perpendicular curved surface. The net of a cylinder consists of a parallelogram and two circles.



A net is a two-dimensional representation of a three-dimensional figure that can be folded to form the three-dimensional shape.

1. Discuss with a partner if these nets can be folded into a right circular cylinder. Circle the nets that represent a net of a right circular cylinder.



12.1.2 GUIDED INSTRUCTION: DRAWING THE NET OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Think-Pair-Share

This strategy can be used on questions 1, 2, and 3.

- Students should work on each question individually and record answers.
- Students will pair with their partner or group members to discuss solutions and provide constructive critiques.
- The teacher will select a few groups to share their solutions and methods.

ENRICHMENT

- Why do you think it is possible to have more than one net for a cylinder?
- How are the nets you choose the same?
- How are they different?

12.1.2 GUIDED INSTRUCTION: DRAWING THE NET OF A RIGHT CIRCULAR CYLINDER (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider providing students with a visual of a net of a right circular cylinder. Allow students to hold the right circular cylinder then unfold it to see the net. Have students describe the image of the net before drawing it.

ENRICHMENT

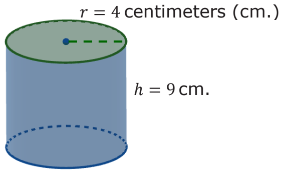
- What do you notice about the location of the height of the cylinder on its net?
- What do you wonder about the length of the curved surface?

TEACHER MOVES

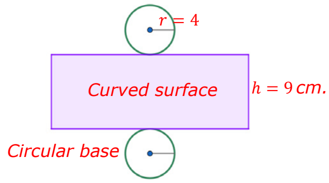
Consider engaging the class in a discussion about radius and diameter to clarify misconceptions.

- How is question 2 different from question 3?
- If given the diameter, how can the radius be found?

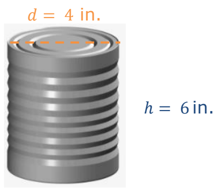
2. Draw the net of the right circular cylinder shown. Label the net with its dimensions. Then, using the terms in the word bank, label its parts.

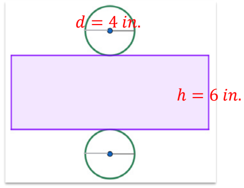


Parts of a Right Cylinder
Circular base
Curved surface



3. Tin cans are used for packaging many things, including various types of foods. The tin can shown has a diameter of 4 inches (in.) and a height of 6 in. Draw the net of the tin can. Label the net with its dimensions.





12.1.3 Exploration: Relationship Between Circumference and Lateral Area of a Right Circular Cylinder

The perpendicular curved surface that connects the circular bases of the right circular cylinder is also called the ***lateral area*** of the cylinder.

1. The diagrams show a circle with a diameter of 4.9 units. The circle is then rolled along the ruler so that the circumference (represented in green) is "peeled" off.

Diagram 1

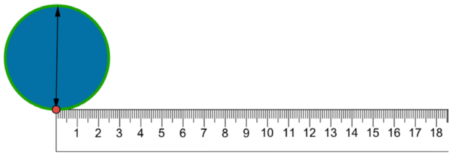


Diagram 2

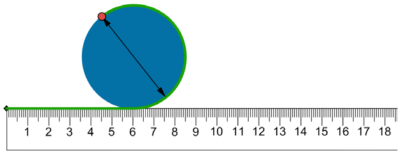
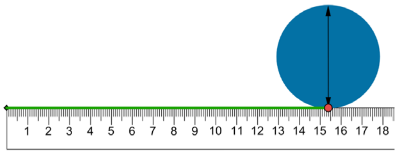


Diagram 3

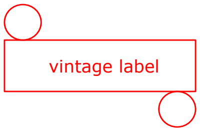


If the circle represents a circular base of a right circular cylinder, discuss with your partner how the diagrams show the relationship between the circumference of the circle and the length of the lateral area of the cylinder. Write a summary of the discussion.

Answers will vary. My partner and I discussed how the "peeled" off circumference from the diagram creates the length of the rectangle for lateral area, which makes the circumference and the length equal to each other.

2. Willard's grandmother collects vintage labels from canned goods. The label shown is from a can of squash.

a. Discuss with your partner how the label relates to the net of a cylinder. Draw a net of the can, labeling the part of the net that represents the vintage label.



12.1.3 EXPLORATION: RELATIONSHIP BETWEEN CIRCUMFERENCE AND LATERAL AREA OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Have different students explain to the class the shape they think the toilet paper roll will be if cut along the dotted line and laid out flat. If toilet paper rolls are not available, students can use index cards or sheets of paper instead.

DIFFERENTIATE INSTRUCTION

For students who might struggle identifying the type of shape, consider having students describe the shape verbally or drawing the shape, specifically making note of the circular base.

TEACHER MOVES

Students can explore the relationship between the circumference of the circular base and the length of the lateral area of a cylinder using the images shown.

TEACHER MOVES

After the toilet paper roll activity, consider showing the students the applet that models how the net is formed from the cylinder. Students will be able to see how the height and radius affect the net. Students can individually engage with the applet to manipulate the model, or you can project the applet for the class to view and ask questions.

ENRICHMENT

Consider prompting questions for students to make sense of the sentence frames provided as a check for understanding.

- How is the net created from the cylinder?
- Describe what the net of a cylinder looks like.
- How is the circumference of the base related to the lateral area?

12.1.3 EXPLORATION: RELATIONSHIP BETWEEN CIRCUMFERENCE AND LATERAL AREA OF A RIGHT CIRCULAR CYLINDER (CONTINUED)

b. Complete the statement to describe how the length of the label is related to the circular base when the label is wrapped around the can.

The length of the label is equal to the

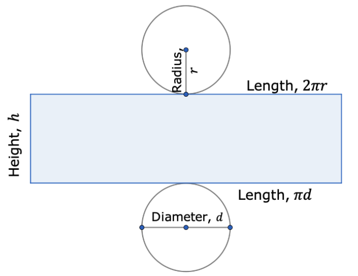
- ○ area
- ○ circumference
- ○ diameter
- ○ radius

of the circular base.

12.1.4 Guided Instruction: Surface Area of a Right Circular Cylinder

The ***surface area of a right cylinder*** is the total area of the two congruent, parallel circular bases and the rectangular curved surface. Surface area is measured in square units.

1. A net of a right circular cylinder is shown with its labeled dimensions.



   a. Discuss with your partner why there are two labels for the length of the rectangle.
   Answers will vary. There are two labels for the length of the rectangle because it is equal to the circumference of the circular base, and there is more than one way to find circumference.

   b. Select all of the formulas that can be used to find the area of 1 of the circular bases and the area of the rectangle.

Area of 1 of the Circular Bases		Area of the Rectangle	
☐ πd^2	☐ $2\pi r \cdot h$	☐ $2 \cdot \pi \cdot d$	☐ πr^2
☐ πr^2	☐ $\pi\left(\frac{1}{2}d\right)^2$	☐ $2\pi r \cdot h$	☐ $2\pi r$
☐ $\pi r \cdot h$	☐ $2\pi r$	☐ πd	☐ $h \cdot \pi \cdot d$

12.1.4 GUIDED INSTRUCTION: SURFACE AREA OF A RIGHT CIRCULAR CYLINDER

TEACHER MOVES

Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about the image of the net of the right circular cylinder. This will allow students the chance to make observations before answering the questions.

DIFFERENTIATE INSTRUCTION

For students who may struggle with the multiple formulas, consider having students sketch two nets of cylinders where one has a radius, and one has a diameter. This will provide students with a visualization of when to use which formula.

12.1.4 GUIDED INSTRUCTION: SURFACE AREA OF A RIGHT CIRCULAR CYLINDER (CONTINUED)

TEACHER MOVES

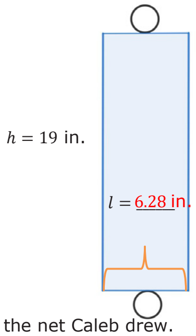
Consider having students explain the steps they took to get their answers. This will help auditory learners to hear the steps from different perspectives.

QUESTIONING

Students who finish early could extend their thinking by answering the following challenge question:

- If Caleb has a second dowel with a height of 12 in. and a radius of 1.5 in., which dowel will have a larger surface area? (The second one has a surface area of 127.17 in^2.)

2. For an art project, Caleb will use a new dowel that measures 2 inches (in.) in diameter and 19 in. in height. To find the surface area of the dowel, Caleb drew the net below. Use 3.14 as π.



   a. Find the missing value in the net Caleb drew. Include units.
   $l = 6.28\ in$

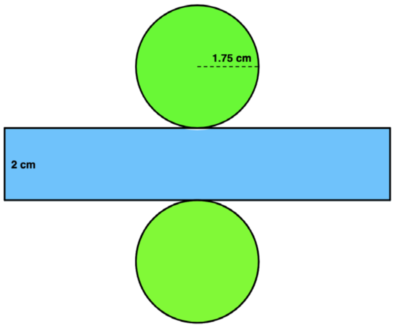
   b. What is the area of one of the circular bases? Include units.
   $3.14\ in^2$

   c. What is the circumference of one of the circular bases? Include units.
   6.28 *in.*

   d. Before Caleb can use the new dowel, he must first remove the tight plastic casing the entire dowel is enclosed in. How much plastic is the dowel enclosed in? Include units.
   $125.6\ in^2$

12.1.5 Wrap-Up: Ticket Out the Door

1. The net of a right circular cylinder is shown.



Use the net to find the surface area of the right circular cylinder. Use $\frac{22}{7}$ for π and include units.

$41.25\ in^2$

12.1.5 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

For students who might struggle finding the surface area, consider having students break down the surface area by finding the area of the circular base and the rectangle first, then combining them to find the surface area.

Alternatively, if students have toilet paper rolls or index cards, prompt them to label these objects with the dimensions provided for an additional entry point.